

Pell Equations

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Introduction:

A pell equation is an equation of the form $x^2 - my^2 = 1$. For example, the most well known pell equation is $x^2 - 2y^2 = 1$. Pell-Type equations are equations of the form $x^2 - my^2 = z$ for some integers z and m . In this article, I will discuss how to find the only solutions to pell equations and the pell-type equation $x^2 - my^2 = 1$, then I will give an example of a problem being solved using Pell Equations.

Pell Equations: $x^2 - my^2 = 1$

A pell equation for this form is $x^2 - 6y^2 = 1$. The solution with the lowest (x, y) is the trivial solution, which in this case is when $x = 5$ and $y = 2$. Therefore when you factor this using difference of squares, we get $(5 - 2\sqrt{6})(5 + 2\sqrt{6}) = 1$. When we square both terms of this equation, we get $(49 - 20\sqrt{6})(49 + 20\sqrt{6}) = 1^2 = 1$ therefore $x = 49, y = 20$ work also. Cubing, taking to the fourth, and in general taking to the n th give the solutions to this pell equation because in the expansion of $(5 - \sqrt{6})^n$, we get all terms with a $\sqrt{6}$ but no 6 will give negative and all terms with a 6 but no $\sqrt{6}$ will give positive.

Pell-Type Equations: $x^2 - my^2 = -1$

A pell equation for this form is $x^2 - 5y^2 = -1$. Then, we have $5y^2 - x^2 = 1$. The trivial solution is $y = 1, x = 2$. Therefore $(\sqrt{5} + 2)(\sqrt{5} - 2) = 1$. When we take this to an even power, we get $(7 + 4\sqrt{5})(7 - 4\sqrt{5})$, however, we want the number to be of the form $(\sqrt{5}y + x)(\sqrt{5}y - x)$ not $(x + \sqrt{5}y)(x - \sqrt{5}y)$. When we cube, we get $(\sqrt{5} + 2)^{2k+1}$ will be in the form $(\sqrt{5}y + x)$ and $(\sqrt{5} - 2)^{2k+1}$ which will have all the $\sqrt{5}$ terms having a odd exponent, therefore the terms with (-2) in this case will be to an even power, giving the $\sqrt{5}$ terms being positive, and the terms with $\sqrt{5}$ being to an even exponent will give (-2) to an odd exponent, therefore all the constant terms are negative. Therefore we get this being of the form $(x\sqrt{5} - y)$, and these cases work. When we have $(\sqrt{5} - 2)^{2k}$, though, we get all the $\sqrt{5}$ terms having an odd exponent therefore the (-2) terms have an odd exponent, giving us negative, and when $\sqrt{5}$ has an even exponent we have (-2) terms having an even exponent, giving us positive, and instead we get $(y - x\sqrt{5})$. Therefore, all terms of the form $(\sqrt{5} + 2)^{2k+1}(\sqrt{5} - 2)^{2k+1}$ will satisfy this pell equation for k being a positive integer.

Finding solutions to $x^2 - my^2 = \pm 1$: If you have a generalized pell equation of the form $x^2 - my^2 = 1$, then find the smallest solution, and any power of this smallest solution will be a solution of the pell equation. To prove this, note that if $(x - \sqrt{m} * y)(x + \sqrt{m} * y) = 1$, then $(x - \sqrt{m} * y)^2(x + \sqrt{m} * y)^2 = 1$ also and in general $(x - \sqrt{m} * y)^n(x + \sqrt{m} * y)^n = 1$. By the Binomial Theorem, we have all terms of the expansion of $(x - \sqrt{m} * y)^n$ with a $\sqrt{m}$ as a factor (but no m) being negative (since $-\sqrt{m} * y$ must be to an odd power), and all terms with a m as a factor instead of a $\sqrt{m}$ will be positive (because we have $(-\sqrt{m} * y)^{2k}$ which is positive). Therefore taking the number to the k th power will give solutions.

For $x^2 - my^2 = -1$, we use $my^2 - x^2 = 1$ giving $(\sqrt{m} * y - x)(\sqrt{m} * y + x) = 1$. All powers to the $2k + 1$ power for k being a positive integer also gives solutions. The proof of this is by noting that all terms with in $(\sqrt{m} * y - x)^{2k+1}$ with only a $\sqrt{m}$ (and no m) in them will have a positive x and all terms with a m (and no $\sqrt{m}$) will have a negative x in them from the Binomial Theorem. Therefore this gives the appropriate form. For $(\sqrt{m} * y - x)^{2k}$ we get all $\sqrt{m}$ but no m terms being negative and all m terms but no $\sqrt{m}$ being positive from the Binomial Theorem. Therefore taking a number to the $2k + 1$ power will give all solutions.

Remark: The ways presented above give all the solutions. The proof of this will not be shown here.

Example 1:

Find the largest positive integer n less than 1000 satisfying the following conditions:

- (i) n^2 can be expressed as the difference of two consecutive cubes;
- (ii) $2n + 79$ is a perfect square.

(2008 AIME II, Problem 15)

Solution:

We need $n^2 = (x+1)^3 - x^3$ and $2n + 79 = m^2$. Therefore $n^2 = 3x^2 + 3x + 1$ and $2n + 79 = m^2$. Start with solving $n^2 = 3x^2 + 3x + 1$. Note that this implies $3x^2 + 3x + (1 - n^2) = 0$ or $x = \frac{-3 \pm \sqrt{9 - 12(1 - n^2)}}{6}$ which implies $x = \frac{-3 \pm \sqrt{12n^2 - 3}}{6}$. Note that we need $\sqrt{12n^2 - 3}$ to be divisible by 3 and 1 (mod 2). For $\sqrt{12n^2 - 3}$ to be an integer, since $12n^2 - 3 \equiv 0 \pmod{3}$, it must be $\equiv 0 \pmod{3}$. Also $12n^2 - 3 \equiv 1 \pmod{2}$ gives us $\sqrt{12n^2 - 3} \equiv 1 \pmod{2}$. Therefore both of these are true and we only account for $\sqrt{12n^2 - 3}$ being an integer. Therefore $\sqrt{3} \cdot \sqrt{4n^2 - 1}$ must be an integer. From this, $4n^2 - 1$ must equal $3z^2$ for $z \in \mathbb{N}$. Therefore $4n^2 - 3z^2 = 1$.

To solve this, we use the same methods we used before for pell equations. The trivial solution is $(1, 1)$ therefore we use $(2 * 1 + \sqrt{3})(2 * 1 - \sqrt{3})$. When we take this to an even power, we are going to get the integer term of $(2 + \sqrt{3})^{2k}$ having all terms with powers of 2 besides for $\sqrt{3}^{2k} \equiv 1 \pmod{2}$ so this constant term is 1 (mod 2). But we need $2n$ being equal to it, so obvious contradiction. For $(2 + \sqrt{3})^{2k+1}$, we get all integer terms having a power of 2, and none not having a power of 2 because the only possible integer term without a power of 2 is $\sqrt{3}^{2k+1}$ which isn't an integer. Therefore this is 0(mod 2) and these are the solutions that work.

Therefore all solutions are going to be of the form $n = \frac{(2+\sqrt{3})^{2k+1} + (2-\sqrt{3})^{2k+1}}{4}$ to get all integer parts of the solution (from $2n = \frac{(2+\sqrt{3})^{2k+1} + (2-\sqrt{3})^{2k+1}}{2}$). Therefore $2n + 79$ which must be a perfect square gives us $\frac{(2+\sqrt{3})^{2k+1} + (2-\sqrt{3})^{2k+1}}{2} + 79 = m^2$. Note that $k = 0$ gives $n = 1$ which works ($2n + 79 = 81$). $k = 1$ gives $n = 13$ giving $2n + 79 = 105$ which isn't a perfect square. $k = 2$ gives $n = 181$ therefore $2n + 79 = 441$ which is a perfect square. $k = 3$ gives $n = 2521$, which is too large. Therefore the only n that work is $n = 1, 181$ of which the largest is 181.

Remark:

All of these values can be found using the Binomial Theorem.

Exercise:

Prove that there exist infinitely many integers n such that $n, n + 1, n + 2$ are each the sum of the squares of two integers. [Example: $0 = 0^2 + 0^2$, $1 = 0^2 + 1^2$, $2 = 1^2 + 1^2$.] (2000 Putnam A2)